



FEATURE USER GUIDE

AML Solutions Risk Rating Analytics White Paper

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Background and Introduction

Navaera's AML Compliance Solutions provide a customer risk rating analytical component that enables automated determination of the faceted risk of a particular customer for engaging in financial crime within a financial services organization (FSO). This faceted risk is determined by analyzing both the attributes of the customer (who they are) and how the customer uses the services of the FSO (what they do).

Applying weights and thresholds to a combination of these factors, the component renders a score for a customer. This score is indicative of the faceted risk within an organization of a particular customer's AML Risk that is relative to the FSO's other customers.

This score is referred to as a 'faceted' AML risk score as it is based purely on objective observations; and derives the score value for each customer based on the identified risks of the FSO's services that the customer uses, as well as the attributes such as the occupation, business type, or geographic factors of the customer that the FSO determines to represent AML risk.

The benefits of this process are twofold. First, transaction monitoring events of higher risk customers will have a higher priority for review, and the activity of those customers will be reviewed with greater scrutiny. Second, it can enable a reduction in workload related to reviewing the activity of lower risk customers. Additionally, once faceted risk rating is configured the organization will be able to methodically rate each customer for AML risk and produce reports to auditors and regulators on the customers identified as high risk, as well as on their related transactions.

As each FSO provides their customers different services, and also caters to different types of customers, the AML Risk Rating component within Navaera's AML Solutions is not designed to provide a 'one size fits-all' approach to risk rate customers. Rather, the component must be tailored to each organization to accurately analyze available data and to assign risk scores to customers appropriately.

This white paper discusses AML Risk Rating concepts in general, and provides insight into best practices in configuring the Risk Rating component within Navaera's AML Solutions.

About Faceted Customer Risk Rating

In order to ensure appropriate procedures are in place to protect the FSO from being used as a conduit for money laundering or terrorist financing, we must identify the FSO's services that may be a higher risk for money laundering (Services Risk); and also the properties of the general customer base that distinguish high risk customers from customers which would have lesser risk (Customer Risk Properties).

Once Services Risk and Customer Risk Properties have been identified, appropriate monitoring procedures can be implemented to focus suspicious activity detection to a greater degree on customers that represent higher risk.

Commonly, the term 'risk-based approach' is used to describe a concept of monitoring where greater scrutiny is applied to services and customers which represent greater AML risk. Navaera's AML solutions facilitate this concept by enabling monitoring to scrutinize the activity of customers that have higher risk attributes, and utilize higher risk services.

While it follows that higher risk customers and services should be monitored with greater scrutiny; it is not recommended that a conclusion should be reached based on the same concept to discontinue all monitoring of low risk customers and services.



Considerations for Identifying Customer Risk Properties

In general, three considerations should be made when identifying Customer Risk Properties, which are:

Customer Relationship Stability: As a Customer Risk Property, the age and type of a customer, as well as the behavior patterns of the customer are typically key risk properties. Considerations should include whether the customer has had a long term relationship with the FSO with stable activity and has consistently used the same products for the duration of their relationship.

Geographic Factors: FSO's with greater geographic diversity also have a more diverse AML risk profile. However geographic factors play a role as a Customer Risk Property even for those FSO's with single locations. Geographic factors that should be considered while identifying customer risk properties include whether the FSO or its customer's location exists within geographies that are known to be high intensity drug trafficking, or related financial crime regions. In certain jurisdictions these geographies are identified. Other geographic factors to consider include location specific features such as the proximity to an international border crossing and even geo-demographic factors.

Referential Factors: Certain types of businesses are more likely to engage in financial crimes than others. Likewise individuals within certain occupations are more likely to engage in financial crimes than others including lawyers, accountants, jewelers, or real estate agents. Other individual profile factors should be considered including whether or not the customer is a politically exposed person (regardless of whether they are exposed domestically or internationally).

Constructing a Customer Risk Properties Matrix

Identifying the business types, occupation codes, geographic factors, relationship factors and other miscellaneous factors that represent higher AML risk enables the translation of Customer Risk Properties to the analytics facility within the Navaera AML Solution. An example Customer Risk Properties Matrix is provided in Appendix A.

Considerations for Identifying Services Risk

Assessing Services Risk involves reviewing each service that the FSO provides, and distinguishing those services which are more likely to be used for financial crime. In general, two considerations should be made when identifying Services Risk, which are:

Internal Controls: Internal controls and policies and procedures designed to control the AML risk should be reviewed for each service, however these controls should not be considered to completely preclude the possibility that certain services may be used for financial crime. Within the perspective of determining which services represent higher risk, organizations should first evaluate the risk of each service in the absence of any implemented controls.

Product Specific Risk: Within each line of services provided by an FSO, specific products are designed for specific assets. An asset with high value that can be bought and sold in a short time frame is attractive to money launderers. However, money laundering is not limited to these types of assets.

Constructing a Services Risk Matrix

Considering the juxtaposition between internal controls and product specific risk should be done in the form of a Services Risk Matrix. A complete Services Risk Matrix identifies each service the FSO provides, the products within the service, the identified risk for each service and product, and the justification for associating the risk with a product. Identifying the transaction types associated with a product enables translating the risk of the product into the analytics facility within the Navaera AML Solution. An example Services Risk Matrix is provided in Appendix B

Specific Components of Risk Rating within Navaera AML Solutions

Navaera AML Solutions utilize six sets of configuration values in the faceted customer risk determination process which are described in the following table.

Configuration Values	Configuration Value Description
Customer Risk Properties (Customer Risk Attributes)	Customer Attribute Risk enables assigning risk to a particular customer based on Customer Risk Properties. Completing a Customer Risk Properties Matrix enables configuring the values for Customer Attribute Risk.
Services Risk (Transaction Risk)	Transaction Risk enables assigning risk to a particular customer based on their use of transactions associated with Services Risk.
Investigation Risk	Investigation Risk can be used as a feature to enable assigning additional risk to customers based on the results of investigations performed on their activity in EnView IMF.
Country Transaction Risk	Country Transaction Risk enables assigning risk to customers based on whether or not they transact with high risk jurisdictions.
Country Attribute Risk	Country Attribute Risk enables assigning risk to customers based on their country of origin.
Domestic Jurisdictional Risk	Domestic Jurisdictional Risk enables assigning risk to customers based on their specific location.

Once a Services Risk Matrix and a Customer Risk Properties Matrix have been completed, the remaining task only involves completing a Country Risk Matrix, where the AML risk of different countries is identified. Then, weighting is conducted on each identified risk to control how and when faceted risk scores are assigned to particular customers, and the effect a customer's faceted risk score has on the transaction monitoring process.

Determining Country Risk

A Country Risk Rating Matrix is used to establish the AML risk of a particular country, and is used for risk rating transactions received from and sent to an identified country; as well as risk rating individuals based on their country of origin. Typical country risk rating methodologies involve objectively determining the risk of particular countries based on international reviews of their money laundering controls. Various international bodies provide such reviews and produce annual reports, including:

- Countries subject to sanctions programs decreed by the European Union (EU);
- Countries subject to sanctions programs decreed by the United Nations (UN);
- Non-Cooperative Countries and Territories (NCCT) identified by the Financial Action Task Force (FATF), and jurisdictions formerly on the list;
- Offshore Financial Centers identified by the International Monetary Fund (IMF);
- Worldwide Governance Indicators (Regulatory Quality, Rule of Law, and Control of Corruption) as determined by the World Bank;
- FATF Statements: geographies identified by the Financial Action Task Force (FATF) as having an inadequate AML/CTF system in place; and
- Un-cooperative Tax Havens identified by the Organization for Economic Co-operation and Development (OECD);

An example of a Country Risk Matrix may be found in Appendix C. In the example, indicators have been researched from a series of identified sources and then weighted for each country. Default settings for Country Risk are also available within Navaera's AML Solutions

Aspects of Jurisdictional Risk

Jurisdictional Risk is a process by which a customer's risk score is impacted by where they are located. There are two ways that is done within the risk rating process.

The first way is to identify the risk of financial and other types of crime near the location itself. This is done similarly to the calculation of foreign jurisdiction risk (e.g. by using multiple sources of information to determine, in this case, the risk of a specific postal code).

In Navaera solutions, data feeds are available that use data from sources including Statistics Canada, DEA, FBI, and provide specific risk levels for each postal code in North America.

Another form of assessing jurisdictional risk, for regional financial services organizations, is to calculate how far a customer resides from the branch where they opened their relationship. Specific distance ranges may then be categorized into high, medium, and low (i.e. those customers that live further would be assessed a higher factor, respectively).

Length of Relationship Impact on Risk Score

The length of a relationship with the financial institution is a big factor when discussing risk. The longer a customer maintains a relationship in good standing with an institution; it would represent a lesser likelihood of money laundering or other financial crimes. The Age Logistics option allows you to lower the risk score on a preset schedule by 99% based on the length of the relationship.

Configuring Risk Weights and Ranges

The overall process for assigning a faceted risk score to a customer is shown in the process diagram below.



As shown above, the first process in assigning a faceted risk score to a customer involves determining the customer's faceted risk properties; then we determine their country risk. Lastly, we determine the investigation and jurisdictional risk. Once each of these processes has occurred, each factor is weighted.

This weighting is done for each type of risk on a scale between 0 and 100. Appendix D provides an example of how to specify this configuration. The weighting assigned to a particular type of risk enables juxtaposition of each factor when determining what the composite faceted risk score should be with all factors considered. After weighting has been determined, the process will be able to determine the faceted risk of each customer. Appendix D provides an example of how to specify this configuration.

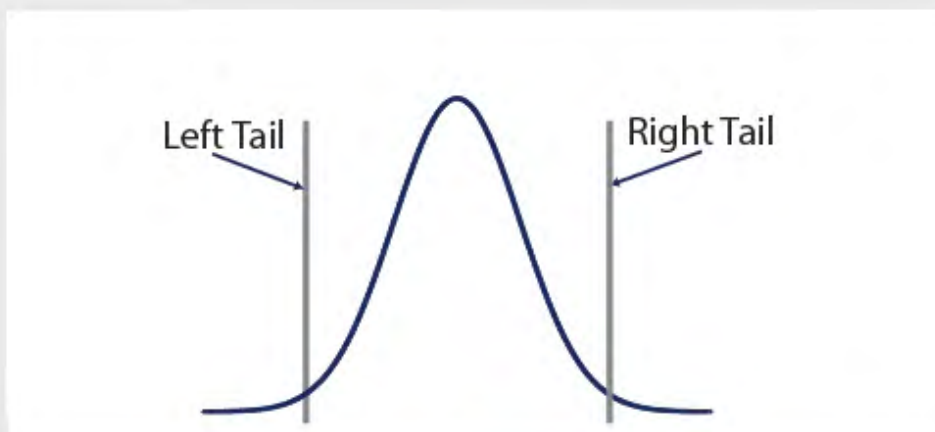
Determining High, Medium, and Low Risk Customers

After all customers have been ranked, it is now possible to specify which scores represent high risk customers, which scores represent medium risk customers, and which scores represent low risk customers. This is done by determining the ranges of scores for each of the three risk levels (e.g. 0-30 represents low risk; 31-60 represents medium risk; and 61 and above represents high risk).

Controlling Transaction Monitoring with Risk Rating

Once the system determines the faceted risk rating for each customer, those assigned risk rating scores may now be used to affect the transaction monitoring process. In order to do this, however, the high, medium, and low risk customer score ranges need to be further divided into three groups. This division occurs by specifying a top percentage of scores as being of higher risk within the risk level, and a bottom percentage is of lower risk within the risk level. We refer to these percentages as tails.

The reason we refer to these percentages as 'tails' is quite straight forward. When analyzing data sets, averages, minimums, and maximums enable us to understand and classify the data within the set. As an example, if we assume that the data within a data set is naturally distributed (e.g. a predictable number of data points are to the right and left of the average value, and we plot that data, a graph called a 'normal curve' is rendered similar to the graph shown below.



The line drawn to the left of the graph indicates a percentage of the data points beyond the identified left tail; and the line drawn to the right of the graph indicates a percentage of the data points beyond the identified right tail, classifying records within the data set into three groups. Likewise for each risk level, high, medium, and low risk, identifying the percentages of those records where the right and left tails exist separates each risk level into three groups.

Applying Risk Rating Considerations to Transaction Monitoring

Once the left and right tails have been identified for each level of risk (high, medium, and low), an 'Event Score Effect' may be configured to adjust the scores of EnView IMF Events from the transaction monitoring process.

In the absence of using customer risk rating to adjust the scores of transaction monitoring Events, each Event is scored only in terms of the observed risk of the detected pattern; typically this only involves basing the score from the number of involved transactions detected that meet the criteria of the pattern. This score is referred to as the 'raw' Event score.

However with the effect of risk rating, the raw score may either be decremented or incremented based on the customer's faceted risk score. To configure this for each of the nine configured groups, a 'multiplier' needs to be identified. This multiplier determines how the raw score is affected by which of the nine groups the customer belongs to. In Appendix D, a matrix has been provided as a framework to determine the multipliers for each of the nine groups.

Conclusion and Purpose

The process of determining faceted customer risk requires the organization to review the properties of its customer base, and the services it provides. While often times this process requires some tedious work and documentation, the introspective review required during this process enables a better understanding of the organization's AML risk profile at a detailed level. At the conclusion of the process each customer within the FSO is rated for AML risk automatically, based on justified factors, and this risk rating is reportable and auditable.

Appendix A: Example Customer Risk Properties Matrix

The Customer Risk Properties Matrix below is an example which may be used to identify various risk properties, and the associated AML Risk Rating for each property. Associated referential data fields should also be identified to enable translating the product risk into the analytics facility within the Navaera AML Solution. It is also a best practice when constructing a Customer Risk Properties Matrix to document the justification for assigning the risk rating to the Customer Risk Property. It should be noted that the matrix below is an example only. Each FSO should construct a matrix based on available data and the types of customers to which the FSO provides services.

Customer Risk Property	AML Risk Rating (0-100)	Risk Rating Justification	Associated Referential Data Fields
Business Entity Types			
Accountants			
Amusement Parks			
Art Dealers			
Automatic Teller Machines (ATM)			
Attorneys			
Auctions			
Automobile Clubs with Traveler's Checks Services			
Beauty/Styling Salons			
Bingo, Gambling			
Card Clubs			
Casinos			
Charter Transports on Boats			
Charity, Foundation, Etc.			

Customer Risk Property	AML Risk Rating (0-100)	Risk Rating Justification	Associated Referential Data Fields
Business Entity Types			
Charter Transport on Buses			
Charter Transport on Airplanes			
Credit Repair Services			
Check Cashing Businesses			
Convenience Stores, Delis, Markets and Supermarkets			
Cruise Lines			
Discotheques			
Doctors			
Embassies and Other Foreign Missions			
Flower Importers			
Foreign Banks			
Foreign Money Exchangers			
Gas Stations			
Hotels/Motels			
Investment Advisors			
Jewelry Stores			
Leather Importers			
Liquor Stores			
Mail Order Houses			

Customer Risk Property	AML Risk Rating (0-100)	Risk Rating Justification	Associated Referential Data Fields
Business Entity Types			
Merchant Wholesalers			
Money Exchanges			
Mortgage and Non Mortgage Loan Brokers			
Off track Betting Agencies			
Pager and Mobile Phone Stores			
Parking Lots and Parking Garages			
Pawn Shops			
Pharmacies and Drugstores, Beauty Supplies Stores			
Precious Metal and Stones Merchant Wholesalers			
Real Estate Brokers			
Real Estate Companies			
Restaurants, Bars			
Sales and Exchange of Money Orders			
Sales and Purchases of Motor Vehicles (Airplanes/RV's)			
Sales and Purchases of Motor Vehicles (Equipment)			
Stock brokers, Securities Dealers			
Taxi, Limousine Companies, Services			
Telegraph Companies			
Title Study Companies			

Customer Risk Property	AML Risk Rating (0-100)	Risk Rating Justification	Associated Referential Data Fields
Business Entity Types			
Tobacco Stores			
Vending Machine Operators			
Video Sales/Rental			
Wire Transfer of Funds Businesses			
Individual occupation types			
Management Occupations			
Business and Financial Operations Occupations			
Computer and Mathematical Occupations			
Architecture and Engineering Occupations			
Life, Physical, and Social Science Occupations			
Restaurants, Bars			
Community and Social Service Occupations			
Legal Occupations			
Education, Training, and Library Occupations			
Arts, Design, Entertainment, Sports, and Media Occupations			
Healthcare Practitioners and Technical Occupations			
Telegraph Companies			
Healthcare Support Occupations			
Protective Service Occupations			

Customer Risk Property	Aml Risk Rating (0-100)	Risk Rating Justification	Associated Referential Data Fields
Individual occupation types			
Personal Care and Service Occupations			
Sales and Related Occupations			
Office and Administrative Support Occupations			
Farming, Fishing, and Forestry Occupations			
Relationship Factors			
Long Term Relationship, Greater than 10 Years			
Medium Term Relationship, Between 2 and 10 Years			
Short Term Relationship, Less than 2 Years			
Miscellaneous Factors			
Marital Status			
Country of Origin			
High Risk Drug Trafficking Area			
High Risk Financial Crimes Area			
Primary Branch Location			

Appendix B: Example Services Risk Matrix

The Services Risk Matrix below is an example matrix which may be used to identify each service the FSO provides, the products within that service, and the associated AML Risk Rating for the product.

Associated transaction types for each product should also be identified to enable translating the product risk into the analytics facility within the Navaera AML Solution. Furthermore, to determine whether or not a particular customer's usage of a product is significant enough to establish that the customer should be associated with use of the service, a number (or count) of transactions should be indicated. This will be used to assign the customer the risk associated with the service if their use of the service exceeds the identified number.

It is also a best practice when constructing a Services Risk Matrix to document the justification for assigning the risk rating to the service. It should be noted that the matrix below is an example only. Each FSO should construct a matrix based on their applicable product and services mix.

Services	AML Risk Rating (0-100)	Risk Rating Justification	Associated Transaction Types	Count of Associated Transactions
ATM Activity				
Domestic ATM Transactions				
Foreign ATM Transactions				
Bond Accounting & Investments				
Bond Sales & Accounting				
Brokered Accounts				
Safekeeping				
Asset/Liability Management				
Investment Portfolio Mgmt				
Federal Funds Programs				
Repurchase Program				
Training/Seminars				

Services	AML Risk Rating (0-100)	Risk Rating Justification	Associated Transaction Types	Count of Associated Transactions
Brokerage & Securities Services				
Annuities				
Brokerage Services				
Investment Advisor				
Professional Asset Management				
Online Securities trading				
Retirement Services				
Correspondent Banking				
Brokered CDs				
DeNovo Bank Services				
Domestic Banks				
Foreign Banks				
Pouch Activity				
Credit Cards				
Domestic Credit Card Transactions				
Foreign Credit Card Transactions				
Electronic Banking Services				
Domestic ACH				
Cross-Border ACH				

Services	AML Risk Rating (0-100)	Risk Rating Justification	Associated Transaction Types	Count of Associated Transactions
Internet Banking				
Online Transfer Service				
Online Bill Payment				
Cash Management Services				
Stored Value Cards				
Payroll Cards				
Vendor Cards				
Gift Cards				
Telephone Banking				
Account Inquiry				
Funds Transfers				
Debit Cards				
Consumer				
Business				
Lending and Mortgage Origination (Separate if Applicable)				
Consumer				
Commercial				
Real Estate				
Residential Lending				
Cross-Border ACH				

Services	AML Risk Rating (0-100)	Risk Rating Justification	Associated Transaction Types	Count of Associated Transactions
Letters of Credit				
Documentary Letters				
Standby Letters				
Retail and Private Banking (Separate if Applicable)				
Demand Deposit				
Negotiable Order of Withdrawal				
Savings				
Money Market				
Time Deposit				
Monetary Instruments				
Official Checks				
Money Orders				
Travelers Checks				
Safe Deposit Boxes				
Trust Services				
Employee Benefit Plans				
Personal Trust				
Estates				
Investment Management				
Corporate Trust				
Custodial Accounts				
Wire Transfer Services				
Domestic Wire Transfers				
Foreign Wire Transfers				

Appendix C: Example Country Risk Matrix

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Australia				1	1	0
Canada				1	1	0
France				1	1	0
New Zealand				1	1	0
Norway				1	1	0
Portugal				1	1	0
Spain				1	1	0
United States				1	1	0
Antigua and Barbuda					2	10
Austria				1	2	10
Barbados				1	2	10
Belgium				1	2	10
Brazil		0.5		2	2	10
Bulgaria				2	2	10
Chile				1	2	10
Comoros					2	10
Costa Rica				2	2	10
Croatia				3	2	10
Czech Republic				2	2	10

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Australia				1	1	0
Canada				1	1	0
France				1	1	0
New Zealand				1	1	0
Norway				1	1	0
Portugal				1	1	0
Spain				1	1	0
United States				1	1	0
Antigua and Barbuda					2	10
Austria				1	2	10
Barbados				1	2	10
Belgium				1	2	10
Brazil		0.5		2	2	10
Bulgaria				2	2	10
Chile				1	2	10
Comoros					2	10
Costa Rica				2	2	10
Croatia				3	2	10
Czech Republic				2	2	10
Denmark				1	1	10
Dominica		0.5			2	10
El Salvador				2	2	10

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Estonia				1	2	10
Finland				1	2	10
Germany				1	2	10
Greece				2	2	10
Hong Kong				1	2	10
Iceland				1	2	10
Ireland	0.5			1	2	10
Israel				1	1	10
Italy				2	1	10
Jamaica				2	2	10
Japan				1	2	10
Korea, South				2	2	10
Latvia				2	2	10
Lithuania				2	2	10
Luxembourg				1	2	10
Malta				1	2	10
Netherlands				1	2	10
Oman				1	3	10
Peru				3	1	10
Qatar				2	2	10
Saint Kitts and Nevis (UK)	0.5				1	10
Singapore				1	2	10
Slovakia				2	2	10

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Slovenia				1	2	10
South Africa				2	2	10
Sweden				1	2	10
Switzerland				1	2	10
Thailand				2	1	10
Trinidad and Tobago				2	2	10
United Arab Emirates				1	2	10
United Kingdom				1	2	10
Uruguay				2	2	10
Albania				4	2	30
Argentina				4	2	30
Bhutan						30
Central African Republic						30
Greenland (Denmark)						30
Guam (US)						30
Honduras				4	2	30
Kiribati						30
Macedonia				4	2	30
Martinique (France)						30
Mayotte (France)						30
Montenegro				4	2	30
Mozambique				4	2	30

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
American Samoa (US)						30
Armenia				3	3	30
Bahrain				2	3	30
Benin				3	3	30
Bosnia and Herzegovina				3	2	30
Botswana				2	3	30
Brunei					3	30
China				3	3	30
Christmas Island (Australia)						30
Cocos (Keeling) Islands (Australia)						30
Djibouti					3	30
Dominican Republic				3	3	30
Egypt	0.5			3	2	30
Falkland Islands (Islas Malvinas) (UK)						30
Faroe Islands (Denmark)						30
Fiji				2	3	30
French Guiana (France)						30
French Polynesia (France)						30
Ghana				3	3	30
Grenada	0.5				2	30

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Guadeloupe (France)						30
Hungary	0.5			2	2	30
India				3	2	30
Jordan				2	3	30
Kuwait				2	3	30
Marshall Islands	0.5				2	30
Mauritius				2	3	30
Mexico				3	2	30
Micronesia, Federated States of					3	30
Moldova				3	2	30
Namibia				2	4	30
New Caledonia (France)						30
Norfolk Island (Australia)						30
Northern Mariana Islands (US)						30
Pitcairn Islands (UK)						30
Poland				3	2	30
Puerto Rico						30
Reunion (France)						30
Romania				3	2	30
Saint Helena (UK)						30
Saint Lucia (UK)					3	30

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Saint Pierre and Miquelon (France)						30
Saint Vincent and the Grenadines (UK)	0.5				2	30
San Marino					3	30
Saudi Arabia				3	3	30
Senegal				3	3	30
Suriname				3	3	30
Svalbard (Norway)						30
Taiwan				2	3	30
Tanzania				3	3	30
Togo					3	30
Tokelau (New Zealand)						30
Tonga					3	30
Tunisia				2	3	30
Turkey				3	3	30
Tuvalu (UK)						30
Virgin Islands						30
Wallis and Futuna (France)						30
Algeria				4	3	70
Andorra			4			70
Anguilla			4			70

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Aruba			4			70
Azerbaijan				5	3	70
Belize			4	2		70
Bermuda			4			70
Bolivia				4	3	70
British Virgin Islands			4			70
Burkina Faso				3	5	70
Cape Verde					4	70
Congo, Democratic Republic of the				5	3	70
Congo, Republic of the				5	3	70
East Timor					4	70
Ecuador				4	3	70
Eritrea				4	4	70
Gabon				3	5	70
Gambia, The				4	4	70
Gibraltar			4			70
Guatemala	0.5			4	2	70
Guernsey			4			70
Guyana				4	3	70
Haiti				5	3	70
Isle of Man			4			70
Jersey			4			70

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Kazakhstan				4	4	70
Kenya				5	3	70
Laos				3	5	70
Lesotho				3	4	70
Macau			4			70
Madagascar				4	3	70
Malawi				4	4	70
Maldives					4	70
Mali		0.75		3		70
Monaco			4			70
Mongolia				3	4	70
Montserrat			4			70
Morocco				3	4	70
Nepal				4	4	70
Netherlands Antilles			4			70
Nicaragua				4	3	70
Niger				4	3	70
Nigeria	0.75			5	2	70
Niue	0.5				3	70
Palau			4			70
Russia	0.5			4	2	70
Rwanda				3	5	70

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Samoa			4			70
Seychelles			4	2		70
Sierra Leone				4	4	70
Sri Lanka				3	4	70
Swaziland				4	3	70
Turkmenistan				5	3	70
Turks and Caicos Islands			4			70
Uganda				4	4	70
Ukraine	0.5			4	2	70
Uzbekistan				5	2	70
Vanuatu			4			70
Vietnam				4	3	70
Zambia				4	4	70
Afghanistan		0.75		4		90
Angola				5	5	90
Bahamas, The	0.5		4			90
Bangladesh				5	4	90
Belarus				4		90
Burma	High			5		90
Burundi				5	5	90
Cambodia				5	4	90
Cameroon				5	4	90

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Cayman Islands	0.5		4			90
Chad				5	4	90
Colombia		High		2		90
Cook Islands (New Zealand)	0.75		4			90
Cyprus		0.75	4	2		90
Equatorial Guinea				5	4	90
Ethiopia				5	4	90
Georgia		0.75		5		90
Guinea					5	90
Guinea-Bissau					5	90
Indonesia	0.75	0.75		5		90
Iraq				5		90
Kyrgyzstan				5	4	90
Lebanon	0.5	High		3		90
Libya				4		90
Liechtenstein	0.5		4			90
Malaysia		0.75	4	2		90
Mauritania		High				90
Pakistan		0.5		5	3	90
Panama	0.5		4	3		90
Papua New Guinea				5	5	90
Paraguay		0.5		5	3	90
Philippines	0.75	0.75		4		90

Country	FATF	Country Reports on Terrorism	Off Shore Financial Center - IMF Assessment	Corruption Perception Index	International Narcotic Control Strategy Report	Risk Assessment (0-100)
Sao Tome and Principe					5	90
Solomon Islands					5	90
Somalia		0.75		5		90
Syria		High		3		90
Tajikistan				5	4	90
Venezuela		High		5		90
Yemen		High		4		90
Zimbabwe				4		90
Nauru	High					90
Côte d'Ivoire				5		100
Cuba				2		100
Iran				3		100
Korea, North						100
Liberia				5		100
Sudan				5		100

Appendix D: Risk Weighting, Range Levels and Event Score Effect Matrix

In this appendix, each remaining configuration matrix is provided in the sections below.

Risk Weighting Matrix

Each type of risk (e.g. Customer Risk Properties, Services Risk, Investigation Risk, Country Transaction Risk, Country Attribute Risk and Domestic Risk) should be weighted to enable the creation of a composite faceted risk score for each customer. The matrix below displays each of the six risk configuration values. The sum of all of the weights should equal 100.

Risk Type	Weight Scale
Customer Risk Properties	
Services Risk	
Investigation Risk	
Country Transaction Risk	
Country Attribute Risk	
Domestic Risk	
Total	100

Risk Level Range Matrix

Each Risk Level (high, medium, and low) must be identified as a part of the range. The matrix below displays each risk level and enables identification of the start and end ranges for each risk level.

Risk Level	Start Range	End Range
High		
Medium		
Low		

Risk Level Left and Right Tails and Sub-Group Determination Matrix

Once the ranges for each risk level have been established, the tail percentages for those ranges can be determined. The matrix below displays each risk level and enables identification of the left and right tail percentages for each risk level.

Risk Level	Left Tail	Right Tail
High		
Medium		
Low		

Event Score Effect Matrix

The Matrix below enables identification of the multipliers for each sub-group. As each risk level (high, medium and low) is further classified into three sub-groups (a high, medium, and low group within each risk level itself) nine total groups are created that should each have a specified multiplier. High subgroups are the right tail of each risk level; low ranges are the left tail of each risk level; and medium indicates records between each identified tail.

Risk Sub-Group	Multiplier
High/High	
High/Medium	
High/Low	
Medium/High	
Medium/Medium	
Medium/Low	
Low/High	
Low/Medium	
Low/Low	

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